

Lokesh Venkatesh

Email: LOKESH.VENKATESH@STUDENTS.IISERPUNE.AC.IN

Research Interests: **Complex Systems** | **Computational Neuroscience** | **Machine Learning**



EDUCATION

Indian Institute of Science Education and Research (IISER), Pune <i>Integrated BS-MS Natural Sciences — Interdisciplinary coursework across the basic sciences</i>	Nov 2022 — Present CGPA: 7.7/10
Yuvabharathi Public School, Coimbatore <i>AISSCE Grade 12 (CBSE) — Physics, Mathematics, Chemistry, Informatics Practices, English</i>	Jul 2022 GRADE: 92.6%
Yuvabharathi Public School, Coimbatore <i>AISSE Grade 10 (CBSE) — Science, Mathematics</i>	Jul 2020 GRADE: 94.4%

RESEARCH EXPERIENCE

Tumour-immune microenvironment interactions in cancer progression <i>Advisors: Prof. Ram Rup Sarkar, NCL Pune & Dr. Chandrakala Meena, IISER Pune</i>	Nov 2025 — Present
Carrying out fixed-time simulations of an ODE model with further parameter & bifurcation analyses to investigate how tumour-immune microenvironment interactions influence the dynamics of cancer progression.	
Self-supervised Representation Learning for Circadian Time Series <i>Advisor: Dr. Shaon Chakrabarti, NCBS Bangalore</i>	May 2025 — Jul 2025
- Implemented representation learning algorithms like the variational autoencoder architecture on patient actigraphy datasets, followed by clustering, in order to obtain their learned low-dimensional representations. - Analysed the learnt embeddings to identify the respective circadian phases and isolate what specific variables carry the most significance in the multivariate time series.	
Study of the Mitochondrial Genome using Motif Distributions <i>Advisor: Prof. MS Madhusudhan, IISER Pune</i>	Aug 2024 — Apr 2025
- Analysed motif distributions across a database of mitochondrial genomes as a metric to both reconstruct phylogenetic relationships over the course of its evolution, as well as identify genomic translocations. - Also looked at possible regulatory correlations between mitochondrial & nuclear genome promoter sequences.	

ACADEMIC PROJECTS & TRAINING

SCHOOLS & WORKSHOPS

Spins, Games & Networks, 2025 Two-week fully-funded winter school on theoretical & computational approaches to complexity science at IMSc Chennai	Dec 2025
NeuroAI Ten-day interactive online course on principles connecting artificial & biological neural nets by the Neuromatch Academy	Jul 2025
Brain, Dynamics & Computation, 2025 Two-week fully-funded summer school on network science-approaches to computational neuroscience at IMSc Chennai	May 2025

COURSE & READING PROJECTS

Tools for Complexity Science — Reading Project Undertaking a reading project (<i>Advisor: Dr. Chandrakala Meena</i>) to study the various methods & tools, largely based on Stefan Thurner et al's book, used to model complex biological and social systems' phenomena.	Sep 2025 — Present
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RELEVANT COURSEWORK

Data Science — Applied Mathematical Methods, Deep Learning, Image & Video Processing, Mathematical & Statistical Optimisation, Computational & Functional Genomics, Signal Processing, Mathematical Finance

Physics — Classical Mechanics, Statistical Mechanics, Nonlinear Dynamics, Physics of Biological Systems

Mathematics — Multivariable Calculus, Linear Algebra, Probability, Graph Theory, Markov Processes

Biology — Systems Biology, Neurobiology, Bioinformatics, Cellular Biophysics I & II, Animal Behaviour, Ecology & Evolution, Cell Biology, Physiology, Genetics

Humanities & Social Sciences — Science & the World, Philosophy of Science, Cognitive Bases of Science, Science & Society, Public Policy I & II

SKILLS

Computational & Modeling: Numerical simulation of ODE systems, Parameter exploration and bifurcation analysis, Time-series analysis, Representation learning algorithms, Bioinformatics Tools

Scientific Computing: High-performance computing, Large-scale data processing, Pipeline development

Programming: Python (NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Biopython), Julia, Bash, Markdown, $\text{\LaTeX}$

Languages: English, Tamil (bilingual proficiency); Hindi (working proficiency)

SCHOLARSHIPS & AWARDS

INSPIRE's Scholarship for Higher Education (SHE)

INSPIRE-SHE Scholar (4 Lakh INR $\approx$ 4,500 USD) - awarded to top 0.1% (approx. 10,000) students nationwide by the Department of Science and Technology, Government of India

EXTRACURRICULAR & OUTREACH

Academic Support Group, IISER Pune

Aug 2023 — Sep 2024

Senior teaching & logistics volunteer for a student collective mentoring socio-economically underprivileged students

Kalpa — IISER Pune's Student Media House

Mar 2023 — Jan 2024

Member of the student media body's core team that coordinated investigative journalism projects on campus matters

REFERENCES

Prof. M.S. Madhusudhan (IISER Pune)

Email: MADHUSUDHAN@IISERPUNE.AC.IN

Dr. Chandrakala Meena (IISER Pune)

Email: CHANDRAKALA@IISERPUNE.AC.IN

Dr. Shaon Chakrabarti (NCBS Bangalore)

Email: SHAON@NCBS.RES.IN

क्र.: एस.एस.ई./2020/
S.No.: S S E / 2020 /

0511459

रजि. नं.
Regn.No. M120/07144/0112

केन्द्रीय माध्यमिक शिक्षा बोर्ड
CENTRAL BOARD OF SECONDARY EDUCATION
अंक विवरणिका सह प्रमाण पत्र
MARKS STATEMENT CUM CERTIFICATE
माध्यमिक विद्यालय परीक्षा, 2020
SECONDARY SCHOOL EXAMINATION, 2020



यह प्रमाणित किया जाता है कि

This is to certify that **LOKESH VENKATESH**

अनुक्रमांक

Roll No. **20135213**

माता का नाम

Mother's Name **RAMALINGAM SUGANYADEVI**

पिता/संरक्षक का नाम

Father's / Guardian's Name **GOPALAN VENKATESH**

जन्म तिथि

Date of Birth **11/07/2004 11TH JULY TWO THOUSAND FOUR**

विद्यालय

School **55167-YUVABHARATHI PUB SCH SOMAYAPALAYAM COIMBATORE TN**

की शैक्षणिक उपलब्धियां निम्नानुसार हैं has achieved Scholastic Achievements as under :

विषय कोड SUB. CODE	विषय SUBJECT	प्राप्तांक MARKS OBTAINED				स्थितीय ग्रेड POSITIONAL GRADE
		लिखित THEORY	आं. मू. IA/ प्रा. PR.	योग TOTAL	योग (शब्दों में) TOTAL (IN WORDS)	
184	ENGLISH LNG & LIT.	075	019	094	NINETY FOUR	A1
006	TAMIL	076	019	095	NINETY FIVE	A2
041	MATHEMATICS STANDARD	072	020	092	NINETY TWO	A1
086	SCIENCE - THEORY	071	020	091	NINETY ONE	A1
087	SOCIAL SCIENCE	080	020	100	ONE HUNDRED	A1

संक्षिप्तों का अर्थ : Abbreviations

AB : अनुपस्थित Absent

IA : आंतरिक मूल्यांकन Internal Assessment

प्रा./PR. : प्रायोगिक/Practical

ER : अनिवार्य पुनरावृत्ति सभी विषयों में

Essential Repeat in all subjects

दिल्ली Delhi

दिनांक Dated : **15-07-2020**परिणाम Result **PASS**

परीक्षा नियंत्रक

Controller of Examinations

सह-शैक्षणिक उपलब्धियां : सह-शैक्षणिक एवं अनुशासन क्षेत्र में ग्रेडिंग विद्यालय द्वारा अपने स्तर पर बोर्ड द्वारा जारी प्रारूपनुसार प्रदान की जाती है।

Co-Scholastic achievements : Grading for Co-Scholastic and Discipline area is being issued by the school as per format prescribed by the Board.

क्रम संख्या/
S.No.SSCE/2022/

1501318

रजि. नं.
Regn.No.

M222/55167/0006

केन्द्रीय माध्यमिक शिक्षा बोर्ड
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MARKS STATEMENT CUM CERTIFICATE
सीनियर स्कूल सर्टिफिकेट परीक्षा, 2022
SENIOR SCHOOL CERTIFICATE EXAMINATION, 2022



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This is to certify that **LOKESH VENKATESH**

अनुक्रमांक

Roll No. **20660746**

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पिता/संरक्षक का नाम

Father's / Guardian's Name **GOPALAN VENKATESH**

विद्यालय

School **55167 YUVABHARATHI PUB SCH SOMAYAPALAYAM COIMBATORE TN**

की शैक्षणिक उपलब्धियां निम्नानुसार हैं has achieved Scholastic Achievements as under :

विषय कोड SUB. CODE	विषय SUBJECT	प्राप्तांक MARKS OBTAINED				स्थितीय ग्रेड POSITIONAL GRADE
		लिखित THEORY	प्रा./PR. आ.मू./IA	योग TOTAL	योग (शब्दों में) TOTAL (IN WORDS)	
301	ENGLISH CORE	067	020	087	EIGHTY SEVEN	A2
041	MATHEMATICS	075	020	095	NINETY FIVE	A1
042	PHYSICS	066	030	096	NINETY SIX	A1
043	CHEMISTRY	065	030	095	NINETY FIVE	A1
065	INFORMATICS PRAC. (NEW)	060	030	090	NINETY	A2
500	WORK EXPERIENCE					A2
502	HEALTH & PHYSICAL EDUCATION					A1
503	GENERAL STUDIES					A1

परिणाम Result **PASS**

दिल्ली Delhi

दिनांक Dated : **22-07-2022**

परीक्षा नियंत्रक
Controller of Examinations

सह-शैक्षिक उपलब्धियां : सह-शैक्षिक एवम् अनुशासन क्षेत्र में ग्रेडिंग विद्यालय द्वारा अपने स्तर पर बोर्ड द्वारा जारी प्रारूपनुसार प्रदान की जाती है।

Co-Scholastic achievements : Grading for Co-Scholastic and Discipline area is being issued by the school as per format prescribed by the Board.

परिवर्तित विषय कोड में, परीक्षा परिणाम 100 अंकों के अंतर्गत दिया गया है।

In the changed subject code, the test result is given under 100 marks.



Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **1**

Chronological Transcript: AUG 2022

	Course code	Course title	Credits	Grade
1	BI1113	Introductory Biology - I	3	C+
2	BI1123	Experimental Biology	3	B
3	CH1113	Principles of Organic Chemistry	3	B+
4	EC1213	Evolution of Earth and Life	3	B
5	MT1113	Calculus - I	3	A
6	PH1113	Introductory Mechanics	3	A+
7	PH1123	Physics Lab - I	3	B
8	TD1113	Introduction to computing	3	A+
9	TD1123	Academic Communication Skills	3	P

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
Cumulative	27	27	8.0

Date of Approval: 17-Mar-2023

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

This is an electronically generated transcript.



Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **2**

Chronological Transcript: JAN 2023

	Course code	Course title	Credits	Grade
1	BI1213	Introduction to Biomolecules	3	B+
2	CH1213	Principles of Physical Chemistry	3	A
3	CH1223	General Chemistry Practicals I	3	B+
4	EC1223	The Solid Earth	3	A
5	HS1223	Science and Society	3	A+
6	MT1213	Calculus II	3	B+
7	MT1223	Linear Algebra	3	B+
8	PH1213	Introductory Electricity and Magnetism	3	A+

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
Cumulative	51	51	8.4

Date of Approval: 27-Jun-2023

Date of Download: 24-Dec-2025

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Dean, Academics

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Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **3**

Chronological Transcript: AUG 2023

	Course code	Course title	Credits	Grade
1	BI2113	Ecology and Evolution	3	B
2	BI2123	Introduction to Biological Systems (Elective)	3	B+
3	CH2113	Principles of Inorganic Chemistry	3	C
4	EC2113	Introduction to Climate Science	3	C+
5	HS2123	Introduction to HSS	3	B
6	MT2113	Introduction to Probability	3	B+
7	PH2113	Introductory Quantum Physics	3	A
8	PH2123	Mathematical Methods for Physics (Elective)	3	A

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
August 2023	24	24	7.4
Cumulative	75	75	8.0

Date of Approval: 12-Dec-2023

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

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भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान पुणे
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH PUNE
Five year BS-MS Dual Degree Programme
Student's Academic Record

Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **4**

Chronological Transcript: JAN 2024

	Course code	Course title	Credits	Grade
1	BI2213	Cell Biology (E)	3	C
2	BI2223	Physiology (E)	3	C
3	BI2233	Genetics (E)	3	C+
4	MT2223	Real Analysis I (E)	3	B+
5	MT2233	Discrete Structures (E)	3	B
6	PH2213	Classical mechanics (E)	3	C
7	TD2213	Thermodynamics	3	C+
8	TD2223	Data Analysis	3	B+

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
August 2023	24	24	7.4
January 2024	24	24	6.3
Cumulative	99	99	7.6

Date of Approval: 10-May-2024

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

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Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **5**

Chronological Transcript: AUG 2024

	Course code	Course title	Credits	Grade
1	BI3134	Bioinformatics	4	B+
2	BI3144	Cellular Biophysics I	4	A
3	BI3154	Neurobiology - I	4	B
4	DS3154	Applied Mathematical Methods	4	A+
5	MT3154	Graph Theory	4	C+
6	DS3313	Semester Project	3	A

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
August 2023	24	24	7.4
January 2024	24	24	6.3
August 2024	23	23	8.1
Cumulative	122	122	7.7

Date of Approval: 09-Dec-2024

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

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Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **6**

Chronological Transcript: JAN 2025

	Course code	Course title	Credits	Grade
1	BI3234	Animal Behaviour	4	C+
2	DS3284	Topics: Mathematical Finance	4	B
3	HS3453	Public Policy I	4	B
4	PH3234	Statistical Mechanics I	4	C+
5	BI4233	Cellular Biophysics 2	3	B
6	DS3273	Image and video processing with deep learning	3	A

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
August 2023	24	24	7.4
January 2024	24	24	6.3
August 2024	23	23	8.1
January 2025	22	22	6.9
Cumulative	144	144	7.6

Date of Approval:

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

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Name of the student **LOKESH VENKATESH** Joining Semester **AUG 2022**
Roll Number **20221150** Residence Semester **7**

Chronological Transcript: AUG 2025

	Course code	Course title	Credits	Grade
1	DS3324	Markov Chain and its Application	4	B+
2	DS4144	Deep Learning	4	A
3	HS3514	Public Policy II	4	A
4	PH4273	Non-linear Dynamics	3	A
5	SE4123	Science and the world	3	B+

Grade Values: A+ = 10; A = 9, B+ = 8, B = 7; C+ = 6; C = 5; D = 4;
P = pass, but credits not counted towards GPA; I = incomplete, 0 credits; F = fail, 0 credits.

1. The first digit in the course code signifies the year in which student are expected to take that course. Some courses of level 3 may be taken in year 4, and vice-versa.
2. GPA: average grade value, weighed by the credits of the course, on a 10-point scale.
3. The semester GPA is calculated on the basis of the credits registered for.
4. The cumulative GPA is calculated on the basis of the credits completed.

Semester	Credits Registered	Credits Completed	GPA
August 2022	27	27	8.0
January 2023	24	24	8.8
August 2023	24	24	7.4
January 2024	24	24	6.3
August 2024	23	23	8.1
January 2025	22	22	6.9
August 2025	18	18	8.6
Cumulative	162	162	7.7

Date of Approval:

Date of Download: 24-Dec-2025

S. Chaudhury
Dean, Academics

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